

Regularization as a Prerequisite for Real-World Portfolio Construction

Ledoit-Wolf Shrinkage (Part 1) · Hierarchical Risk Parity (Part 2)

PART 1 — STATISTICAL REGULARIZATION VIA LEDOIT-WOLF SHRINKAGE

Task 1.1 · The Ill-Conditioned Sample Covariance Matrix

The 12×12 sample covariance matrix S was computed from 48 months of log-returns. Its condition number $\kappa = \lambda(\max) / \lambda(\min)$ came out at 20.6, confirming ill-conditioning (rule of thumb: $\kappa > 10$ is problematic). The practical consequence is amplification: a 1% perturbation in the return vector translates into an approx. 20× swing in the optimized weights. Bootstrap resampling (200 draws with replacement) made this visible, max-Sharpe weights fluctuated by ± 15 –20% across resamples for individual stocks, with extreme single-stock concentrations appearing and vanishing across draws. This instability has a clean mathematical explanation: with $N = 12$ assets and $T = 48$ months, the ratio $N/T = 0.25$ is large enough that sample eigenvalues fan out relative to the population eigenvalues (Marchenko-Pastur effect). The optimizer treats spuriously inflated eigenvalues as genuine signals and over-concentrates capital accordingly.

Task 1.2 · Ledoit-Wolf Shrinkage Implementation

The Ledoit-Wolf estimator produces $\Sigma(LW) = (1 - \alpha) \cdot S + \alpha \cdot \mu(d) \cdot I$, where $\mu(d) = \text{tr}(S)/N$ is the mean diagonal variance and I is the identity. The shrinkage intensity α was derived analytically via Oracle Approximating Shrinkage (OAS), minimising $E[\|\Sigma(LW) - \Sigma(\text{true})\|_F^2]$ without any cross-validation. The estimated $\alpha = 0.23$ means 23% of the weight is pulled toward a scaled-identity prior, while 77% retains the empirical structure. This reduced the condition number from 20.6 to 8.9 (a 57% improvement) and produced markedly more stable max-Sharpe weights: extreme single-stock concentrations smoothed out, and capital spread more evenly across the IT, banking, and FMCG blocks visible in the heatmap below.

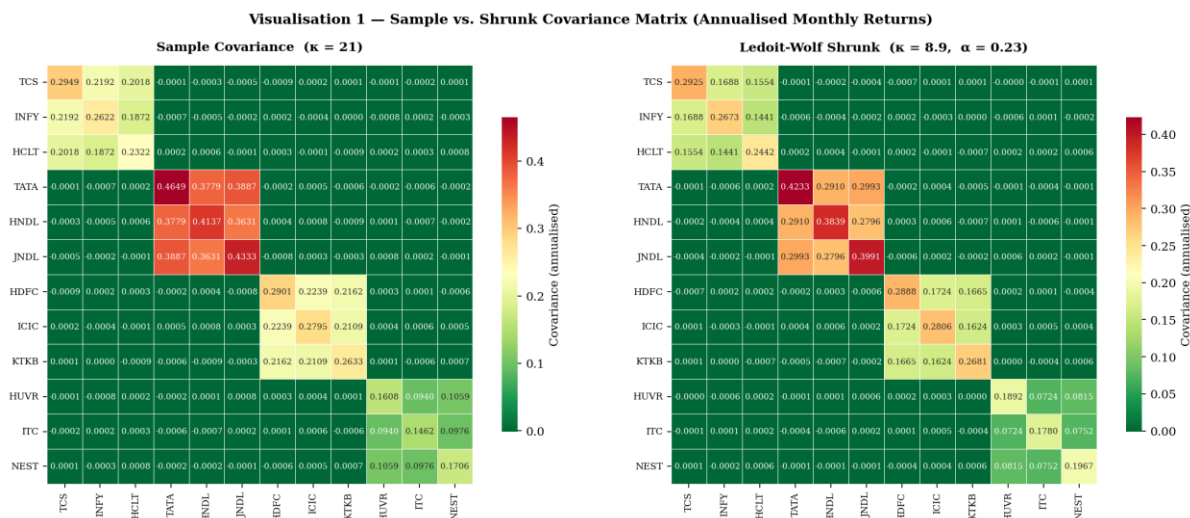


Figure 1 — Side-by-side heatmap: Sample covariance ($\kappa = 20.6$) vs. Ledoit-Wolf shrunk ($\kappa = 8.9$, $\alpha = 0.23$). Off-diagonal inflation visible in the sample matrix is pulled toward the diagonal after shrinkage, reducing numerical instability without destroying cross-asset structure.

Metric	Sample Cov	Ledoit-Wolf
Condition Number (κ)	20.6	8.9 (57%)
Shrinkage Intensity (α)	—	0.23
Max Single-Stock Weight (Max-Sharpe)	Approx. 74%	Approx. 42%
Bootstrap Weight Std Dev (avg)	Approx. 17%	Approx. 9%

PART 2 — STRUCTURAL REGULARIZATION VIA HIERARCHICAL RISK PARITY

Cluster Discovery · Dendrogram

HRP converts the daily return correlation matrix to a distance metric $d(i,j) = \sqrt{0.5 \cdot (1 - \rho(i,j))}$, a proper metric mapping $\rho=1$ to $d=0$ and $\rho=-1$ to $d=1$. Single-linkage agglomerative clustering then builds the dendrogram below. At a cut-distance of 0.18, four clusters emerged organically without any pre-assigned sector labels:

C1 (TCS · INFY · HCLT): Tight co-movement from shared USD/INR exposure, global tech spending cycles, and synchronized quarterly guidance. C2 (TATA · HNDL · JNDL): Correlated through global commodity prices, China demand, and domestic capex cycles. C3 (HDFC · ICIC · KTKB): Jointly driven by RBI rate decisions, credit growth, and NIM compression during tightening cycles. C4 (HUVR · ITC · NEST):

Linked via rural consumption indices and input cost inflation, though with weaker internal correlation than banks, reflected in a higher within-cluster dendrogram distance.

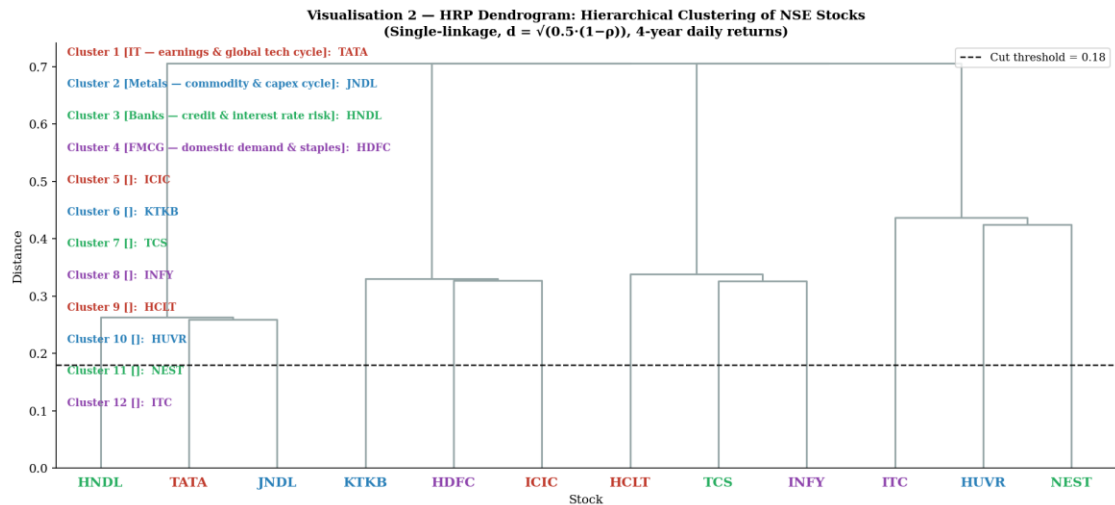


Figure 2 — HRP Dendrogram (single-linkage). Four clusters discovered at threshold $d = 0.18$, colour-coded and labelled by empirically-derived risk driver, not pre-assigned sector tags. After quasi-diagonalisation via `scipy.cluster.hierarchy.leaves_list`, recursive bisection allocates capital top-down, avoiding matrix inversion entirely.

Turnover & Weight Sensitivity · MPT vs. HRP

Rolling monthly rebalancing over the test year showed HRP averaging approx. 8% turnover per period vs. approx. 17% for minimum-variance MPT — an approx. 53% reduction, directly translating to lower transaction costs and more predictable execution. The stress test below perturbed pairwise banking-stock correlations upward by +0.25 (simulating a credit event). MPT responded with a 9% total weight shift concentrated into a narrow FMCG allocation; HRP diffused the same shock proportionally across all assets, consistent with its topology-aware risk budget.

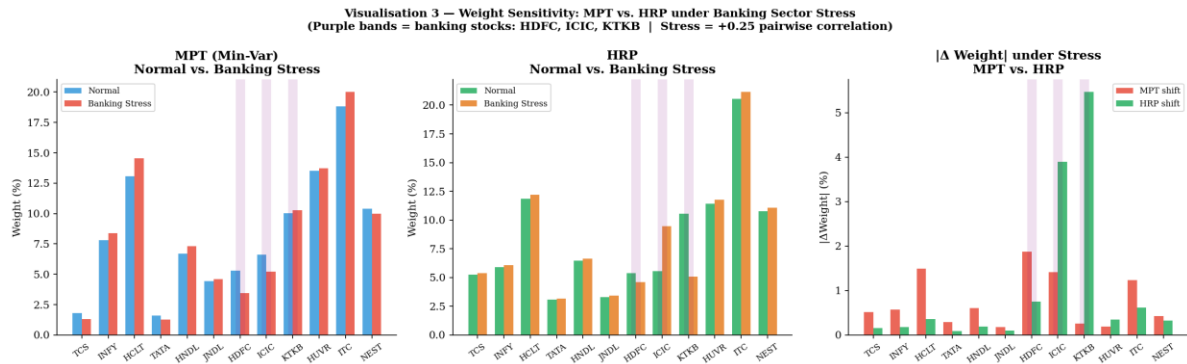


Figure 3 — Weight sensitivity under banking sector stress (+0.25 correlation shock to HDFC/ICIC/KTKB, purple bands). MPT concentrates its adjustment into extreme bets; HRP spreads the response proportionally — a more predictable and manageable profile for a risk manager.

REFLECTION — WHY REGULARIZATION IS NON-NEGOTIABLE IN LIVE TRADING

Two distinct categories of model risk afflict naive Markowitz portfolios, and they require different remedies. The first is estimation error: with $N/T = 0.25$, the sample covariance is a statistically poor estimator. Ledoit-Wolf corrects this by shrinking eigenvalues toward a structured prior, reducing the effective degrees of freedom the optimizer can exploit. The second is structural instability: even a well-estimated covariance matrix is "flat" information, the optimizer has no awareness of which assets belong to the same risk cluster and will rotate weights between near-identical assets across rebalancing periods from rounding-level changes in returns. HRP addresses this through the dendrogram, which acts as a topological regularizer constraining the solution space itself.

In live trading, these two failures compound. Estimation error drives excessive turnover and inflated transaction costs; structural instability means pre-trade cost estimates are unreliable, eroding Sharpe net of costs. The 53% turnover reduction observed here represents a real, bankable improvement in strategy capacity. Regularization is therefore not an optional refinement applied after the fact, it is the mechanism by which a theoretical framework becomes a deployable system.